

Commerce XI Maths App

Theory (Pages 16-18)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 25

Question 1 - MCQ · PG-16 · TH-Q1

An ogive is a

Options

- A. cumulative frequency curve
- B. frequency polygon
- C. bar diagram
- D. pie chart

Correct answer: cumulative frequency curve

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives cumulative frequency curve.

Question 2 - MCQ · PG-16 · TH-Q2

The two types of ogives are

Options

- A. less than ogive and more than ogive
- B. upper and lower ogive
- C. positive and negative ogive
- D. simple and compound ogive

Correct answer: less than ogive and more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than ogive and more than ogive.

Question 3 - MCQ · PG-16 · TH-Q3

For a less than ogive, plot upper class boundaries against

Options

- A. less than cumulative frequencies
- B. class frequencies
- C. more than cumulative frequencies
- D. class marks

Correct answer: less than cumulative frequencies

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than cumulative frequencies.

Question 4 - MCQ · PG-16 · TH-Q4

For a more than ogive, plot lower class boundaries against

Options

- A. more than cumulative frequencies
- B. less than cumulative frequencies
- C. class marks
- D. relative frequencies

Correct answer: more than cumulative frequencies

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than cumulative frequencies.

Question 5 - MCQ · PG-16 · TH-Q5

The curve obtained by joining cumulative-frequency points smoothly is called an

Options

- A. ogive
- B. histogram
- C. frequency polygon
- D. sector diagram

Correct answer: ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ogive.

Question 6 - MCQ · PG-17 · TH-Q6

To locate the median on an ogive, draw a horizontal line from

Options

- A. $N/2$ on the Y-axis
- B. $N/4$ on the Y-axis
- C. $2N$ on the X-axis
- D. N on the X-axis

Correct answer: $N/2$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $N/2$ on the Y-axis.

Question 7 - MCQ · PG-17 · TH-Q7

To locate Q_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(4)$ on the Y-axis
- B. $(i N)/(10)$ on the Y-axis
- C. $(i N)/(100)$ on the Y-axis
- D. $(N)/i$ on the X-axis

Correct answer: $(i N)/(4)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(4)$ on the Y-axis.

Question 8 - MCQ · PG-17 · TH-Q8

To locate D_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(10)$ on the Y-axis
- B. $(i N)/(4)$ on the Y-axis
- C. $(i N)/(100)$ on the Y-axis
- D. $10N/i$ on the X-axis

Correct answer: $(i N)/(10)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(10)$ on the Y-axis.

Question 9 - MCQ · PG-17 · TH-Q9

To locate P_i on an ogive, draw a horizontal line from

Options

- A. $(i N)/(100)$ on the Y-axis
- B. $(i N)/(10)$ on the Y-axis
- C. $(i N)/(4)$ on the Y-axis
- D. $100N/i$ on the X-axis

Correct answer: $(i N)/(100)$ on the Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(100)$ on the Y-axis.

Question 10 - MCQ · PG-17 · TH-Q10

The intersection point of less than and more than ogives locates the

Options

- A. median
- B. mode
- C. mean
- D. range

Correct answer: median

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives median.

Question 11 - MCQ · PG-17 · TH-Q11

Partition values obtained graphically are generally

Options

- A. approximate
- B. always exact
- C. always integers
- D. undefined

Correct answer: approximate

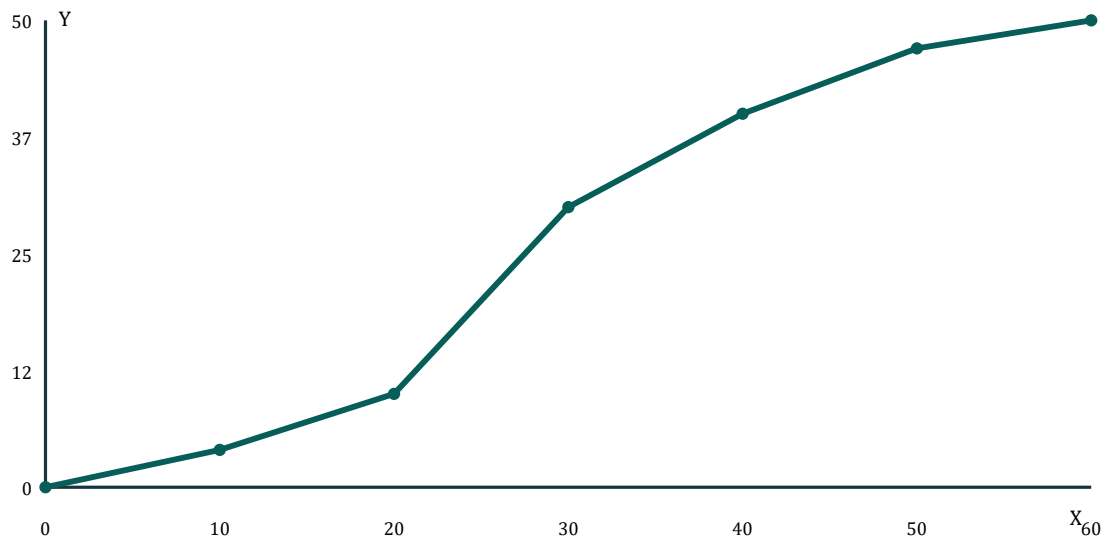
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives approximate.

Question 12 - MCQ · PG-17 · TH-Q12

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	4	6	20	10	7	3

From the ogive shown below, Q_2 is approximately



Options

- A. 27.5
- B. 25
- C. 30
- D. 40

Correct answer: 27.5

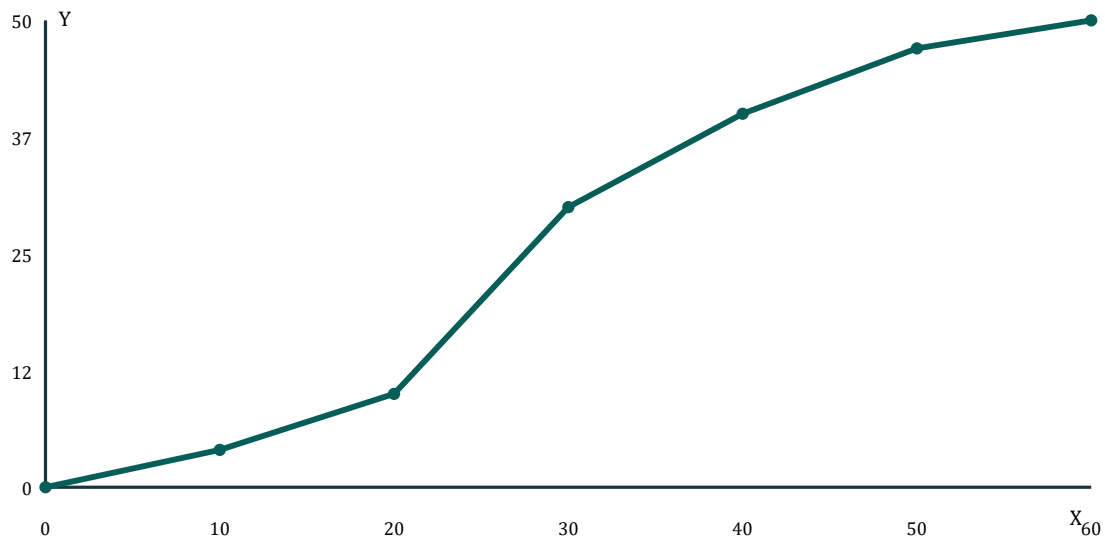
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 27.5.

Question 13 - MCQ · PG-17 · TH-Q13

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	4	6	20	10	7	3

From the ogive shown below, D_8 is approximately



Options

- A. 40
- B. 27.5
- C. 30
- D. 50

Correct answer: 40

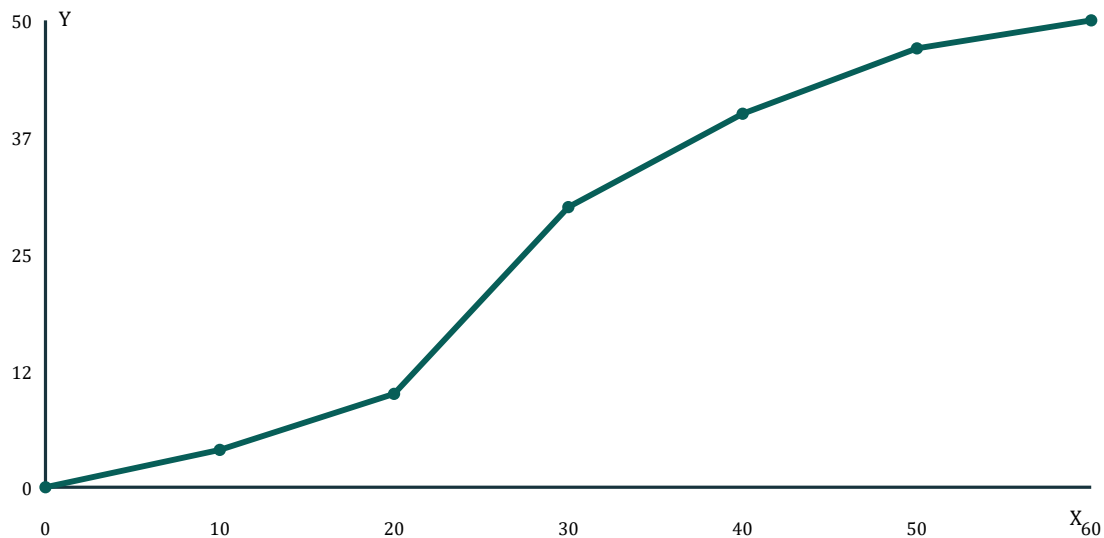
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40.

Question 14 - MCQ · PG-17 · TH-Q14

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	4	6	20	10	7	3

From the ogive shown below, P_{60} is approximately



Options

- A. 30
- B. 27.5
- C. 40
- D. 60

Correct answer: 30

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 30.

Question 15 - MCQ · PG-18 · TH-Q15

Before plotting discontinuous classes 10–14, 15–19, ..., they are converted to

Options

- A. continuous classes 9.5–14.5, 14.5–19.5, ...
- B. class marks only
- C. percentages
- D. descending frequencies

Correct answer: continuous classes 9.5–14.5, 14.5–19.5, ...

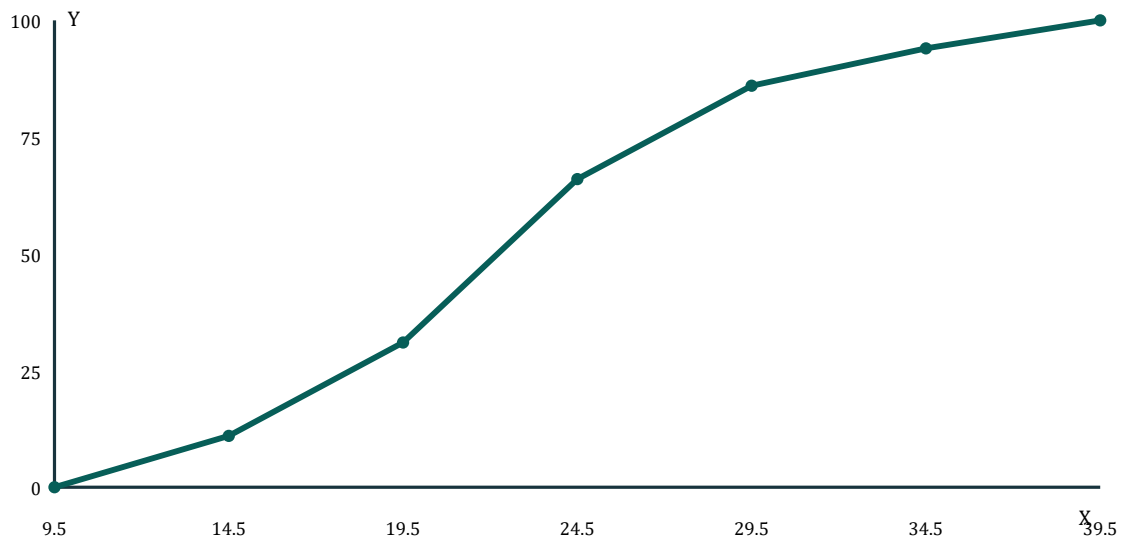
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives continuous classes 9.5–14.5, 14.5–19.5,

Question 16 - MCQ · PG-18 · TH-Q16

Overtime per week (hours)	10-14	15-19	20-24	25-29	30-34	35-39
No. of employees	11	20	35	20	8	6

From the ogive shown below, Q_1 is approximately



Options

- A. 18
- B. 11
- C. 23
- D. 25

Correct answer: 18

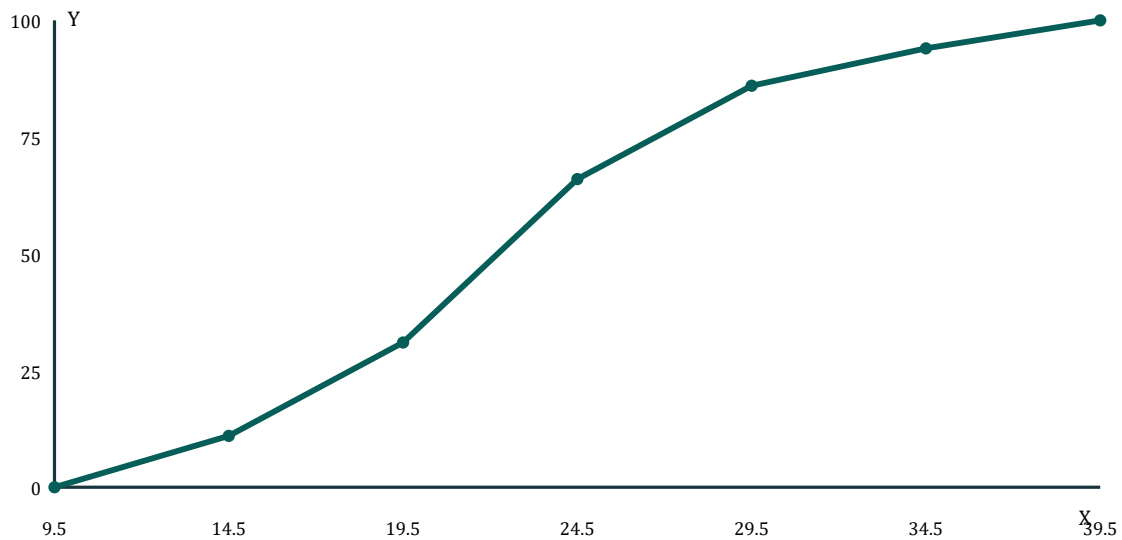
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 18.

Question 17 - MCQ · PG-18 · TH-Q17

Overtime per week (hours)	10-14	15-19	20-24	25-29	30-34	35-39
No. of employees	11	20	35	20	8	6

From the ogive shown below, the median is approximately



Options

- A. 23
- B. 18
- C. 25
- D. 29

Correct answer: 23

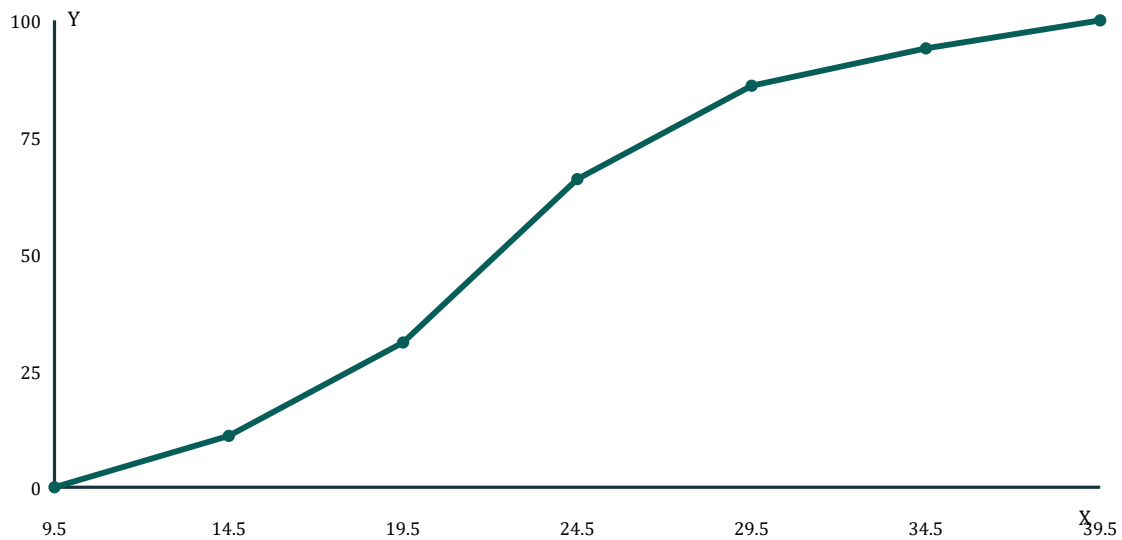
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 23.

Question 18 - MCQ · PG-18 · TH-Q18

Overtime per week (hours)	10-14	15-19	20-24	25-29	30-34	35-39
No. of employees	11	20	35	20	8	6

From the ogive shown below, P_{70} is approximately



Options

- A. 25
- B. 18
- C. 23
- D. 30

Correct answer: 25

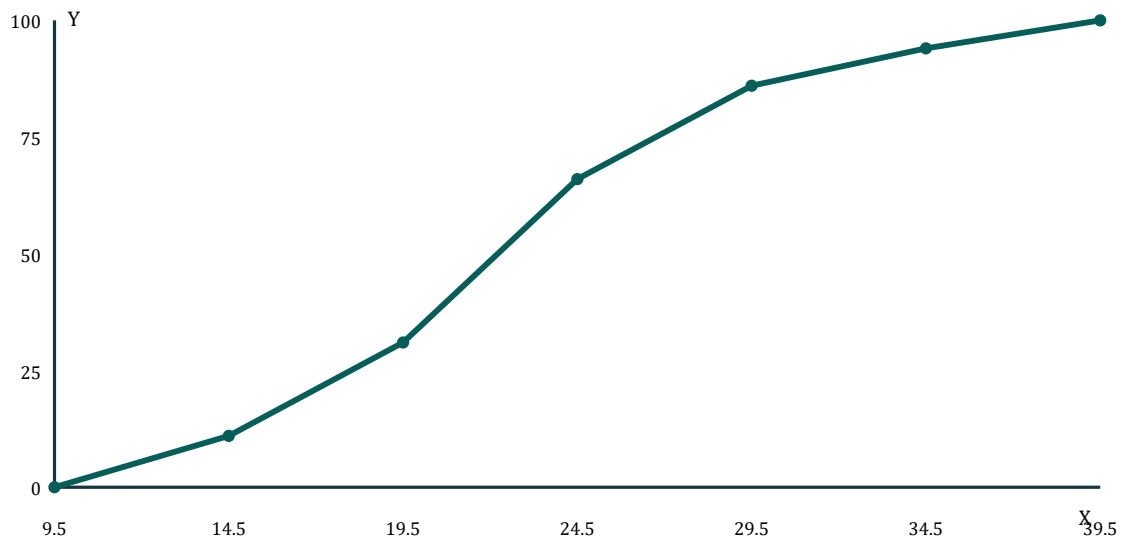
Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 25.

Question 19 - MCQ · PG-18 · TH-Q19

Overtime per week (hours)	10-14	15-19	20-24	25-29	30-34	35-39
No. of employees	11	20	35	20	8	6

From the ogive shown below, employees working less than 11 hours are approximately



Options

- A. 4
- B. 6
- C. 11
- D. 20

Correct answer: 4

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 4.

Question 20 - MCQ · PG-17 · TH-Q20

After meeting the ogive, a perpendicular is drawn to the

Options

- A. X-axis to read the partition value
- B. Y-axis to read total frequency
- C. origin only
- D. class-frequency column

Correct answer: X-axis to read the partition value

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives X-axis to read the partition value.

Question 21 - MCQ · PG-16 · TH-Q21

A zero cumulative frequency is assigned before the first class when drawing a

Options

- A. less than ogive
- B. more than ogive
- C. histogram
- D. pie chart

Correct answer: less than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives less than ogive.

Question 22 - MCQ · PG-16 · TH-Q22

A zero cumulative frequency is assigned after the last class when drawing a

Options

- A. more than ogive
- B. less than ogive
- C. frequency polygon
- D. bar graph

Correct answer: more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than ogive.

Question 23 - MCQ · PG-17 · TH-Q23

The horizontal level used to locate Q_1 is

Options

- A. $N/4$
- B. $N/2$
- C. $(3N)/4$
- D. $N/10$

Correct answer: $N/4$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $N/4$.

Question 24 - MCQ · PG-17 · TH-Q24

The horizontal level used to locate Q_3 is

Options

- A. $(3\text{ N})/(4)$
- B. $N/4$
- C. $N/2$
- D. $(3\text{ N})/(10)$

Correct answer: $(3\text{ N})/(4)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(3\text{ N})/(4)$.

Question 25 - MCQ · PG-18 · TH-Q25

The overtime example can also be solved by drawing a

Options

- A. more than ogive
- B. pie chart only
- C. simple bar chart
- D. scatter plot

Correct answer: more than ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives more than ogive.